Kazakh-British Technical University
School of Information Technology and Engineering

SCADA systems and Industrial networks

DIFFERENT TYPES OF SCADA POINTS

Laboratory work №15

Prepared by: Fazkullin F., Zhanshin T.

Checked by:
PhD, prof. SITE (docent) Samigulin T.I.

Almaty, 2026

CONTENT

Introduction..... 3
1. Tasks..... 4
1.1 Task 1. Configure an analog point..... 4
1.2 Task 2. Configure a status point..... 10
1.3 Task 3. Configure an accumulator point..... 10
Conclusion..... 15
Worksheet..... 16

INTRODUCTION

The primary objective of this laboratory work is to master the fundamental process of database engineering within the Honeywell Experion PKS (Process Knowledge System). In a modern Distributed Control System (DCS), the "Point" is the core logical entity that acts as a functional bridge between physical field signals and the operator's Graphical User Interface (HMI). This laboratory focuses on the systematic creation, addressing, and parameterization of these points using the Quick Builder application.

The first phase of the work is dedicated to the configuration of Analog Points, which are used to monitor continuous variables such as flow, pressure, or temperature. The configuration involves defining Engineering Units (L/sec), scaling raw controller data into a 0–200 range, and establishing a hierarchical alarm system (PVHighHigh, PVHigh, etc.). These settings are critical for process safety, as they allow the system to automatically notify the operator when parameters deviate from safe operating limits.

The second key aspect of the laboratory involves Status Points, which represent the discrete states of multi-position devices such as pumps or valves. By mapping Non-consecutive Bits from the M_CR2 controller, we define logical states such as ON, OFF, FAULT, and TRAVEL. This transforms binary controller signals into human-readable information, enabling the system to not only track equipment operation but also diagnose malfunctions in real-time.

The final part of the laboratory covers the setup of Accumulator Points, designed for totalizing pulse data or flow rates over a specific period (e.g., total liquid volume). Additionally, advanced database management tasks are addressed, including the creation of User-Defined Parameters (USER_VAR, STATUS_COUNT) and the configuration of History Collection. These steps are vital for data archiving, which is necessary for trend analysis, regulatory reporting, and diagnostic troubleshooting within an industrial environment.

1. Tasks

1.1 Task 1. Configure an analog point

I initiated the process by launching the Configuration Studio application and connecting to the system server. In the configuration tree, I navigated to [System Name] → Servers → [My Server Name] and selected the Control Strategy section. From the center frame, I clicked on the Build points link, which opened the Quick Builder application specifically for managing SCADA point entities.

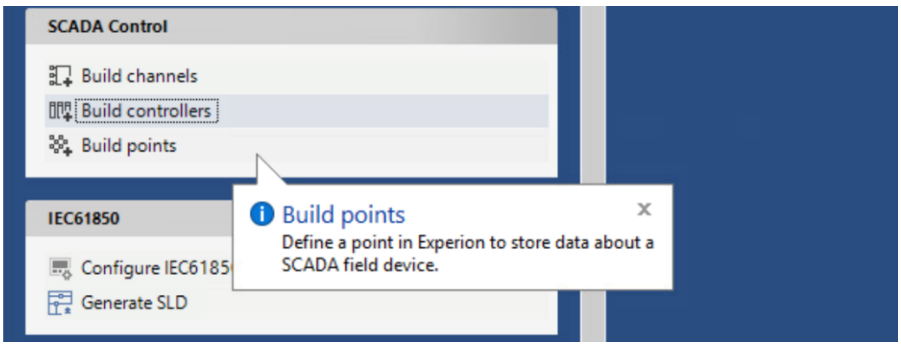


Figure 1 – Build points link

Inside Quick Builder, I clicked the Add icon to create a new database item. From the Type list box, I selected Analog Point to represent a continuous process variable. I assigned the name Analog_Point01 to this point and clicked OK to proceed to the detailed configuration tabs.

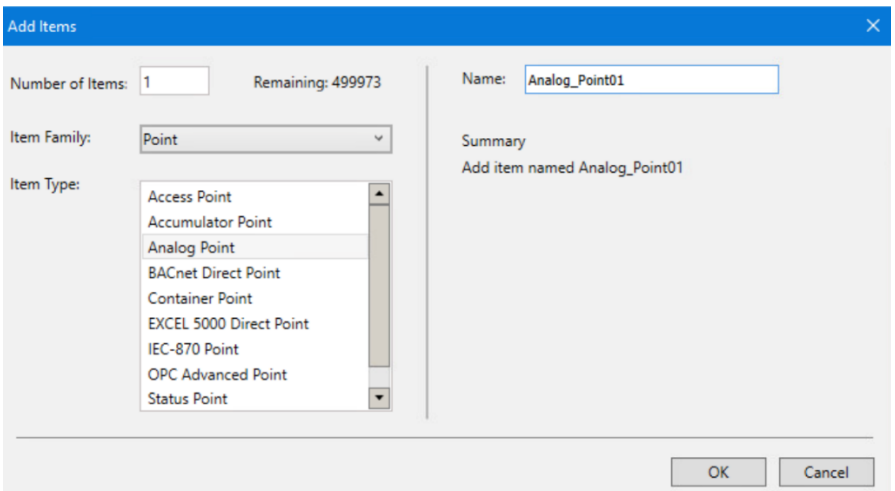


Figure 2 – Adding new analog point

On the Main tab, I defined the point's identity and its place within the system hierarchy. I entered "Distillation flow loop" as the Description and assigned the point to the Distillation_column under the Parent Asset field. To ensure timely data updates, I set the PV Scan Period to 2 seconds and defined the Engineering Unit as "L/sec" to reflect the flow measurement.

I then configured the PV Source Address to link the point to the physical hardware. I set the Address Type to Controller and selected M_CR1 as the target device. In the Location field, I entered the address "1 0", where "1" represents the first address of the Holding Register and "0" specifies the bit number. This setup allows the server to correctly locate and read the raw data from the controller.

To ensure the raw data is correctly scaled, I set the 100% Range Value to 200 and the 0% Range Value to 0. I enabled both Scanning Enabled and Clamp PV checkboxes; the former ensures the server begins refreshing the database from the controller registers immediately upon download, while the latter prevents the Process Variable (PV) from exceeding the defined range. Finally, I set the Drift Deadband to 0, ensuring that even minor changes in the parameter are processed by the system.

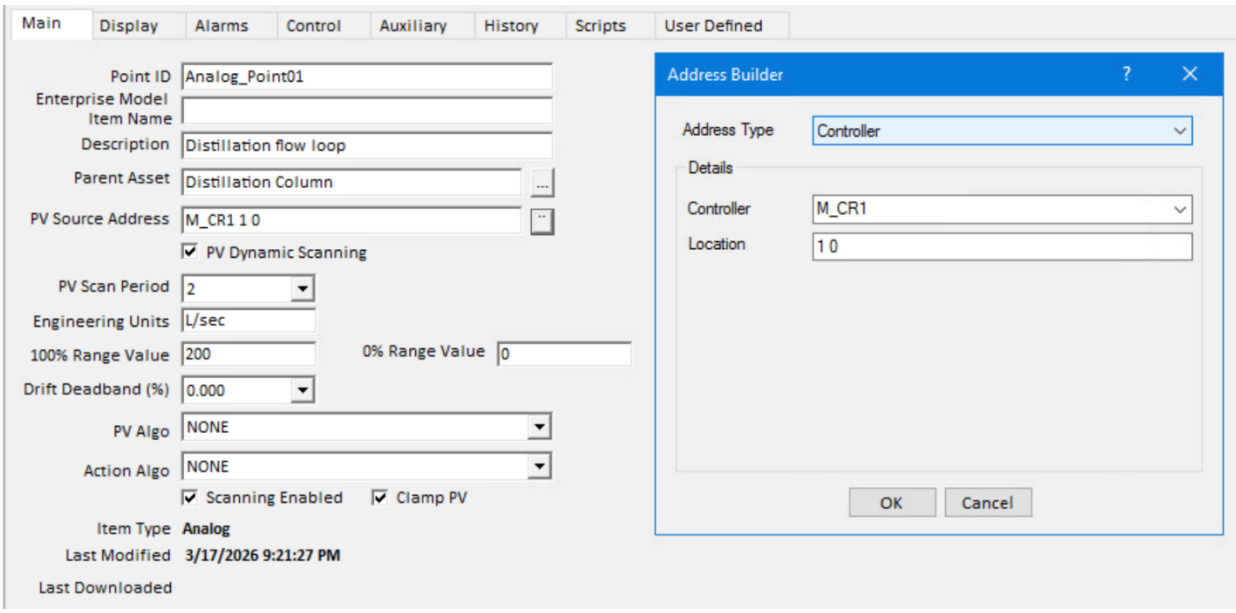


Figure 3 – Analog point configuration (Main tab)

I navigated to Station → click Configure → Alarm Event Management → Message and Point Processing → Point Processing tab to define the operational limits for the point. I configured the High and Low alarm thresholds to ensure the operator is notified if the flow rate deviates from the safe operating range.

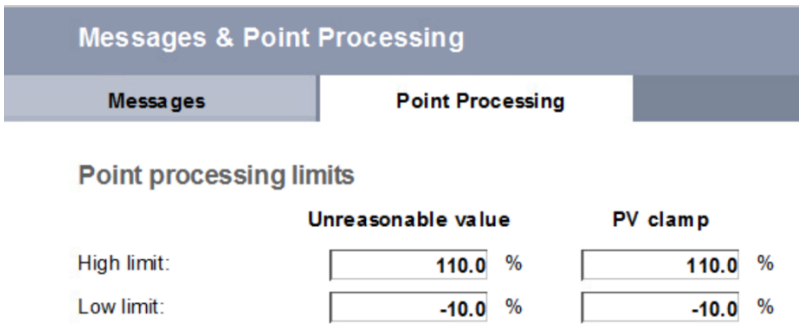


Figure 4 – Point processing limits

On the Display tab, I assigned the point to a specific functional group and trend. I entered the Group number (within the 1 to 16,000 range) and the Trend number (within the 1 to 3,000 range) as shown in the provided documentation. This step is essential for ensuring that the point appears on the correct operator HMI displays and that its real-time data can be visualized in trend charts.

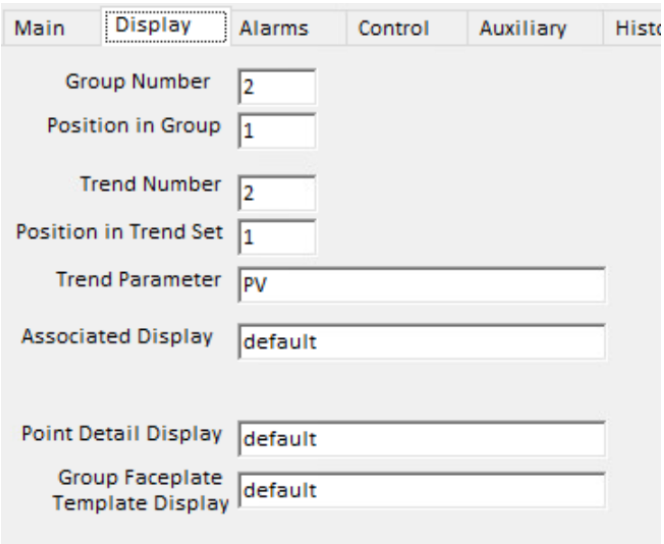


Figure 5 – Analog point configuration (Display tab)

On the Alarms tab, I established the critical operating boundaries for the analog point by defining four distinct thresholds with specific priorities to ensure precise process monitoring. I configured PVHighHigh and PVLowLow with a limit of 180 and 10 respectively, assigning them an Urgent priority to signify critical conditions, while setting the intermediate PVHigh (170) and PVLow (30) thresholds to High and Low priorities. By maintaining the Sub Priority at 0 and the On/Off Delays at 0 seconds, I ensured that the system provides instantaneous alerts to the operator the moment the process variable crosses these safety limits.

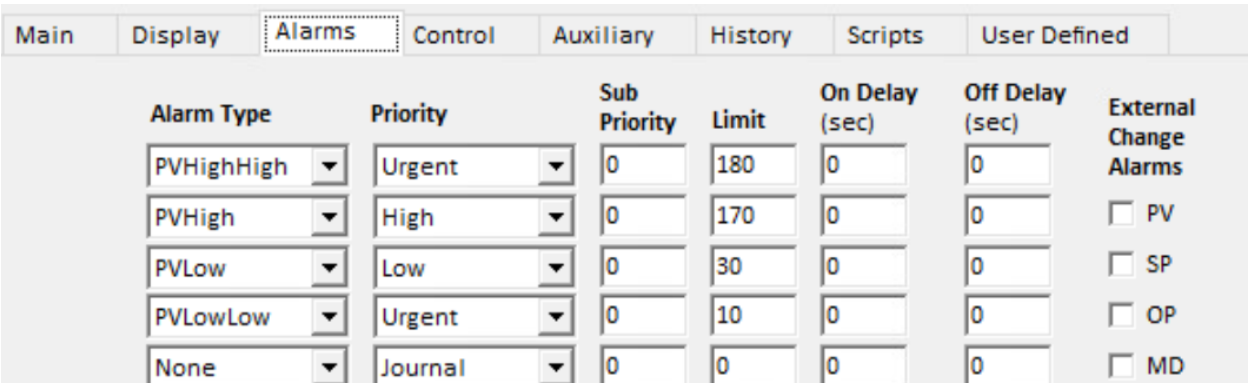


Figure 6 – Analog point configuration (Alarms tab)

I then accessed the Control tab to define how operators interact with the point. I noted that by selecting the "Disable mode checking on output" checkbox, an operator would be able to change the Output (OP) value regardless of the point's

current mode. This bypasses the standard server check that typically requires the point to be in a specific mode (like Manual) before allowing value changes, providing more direct control during the simulation.

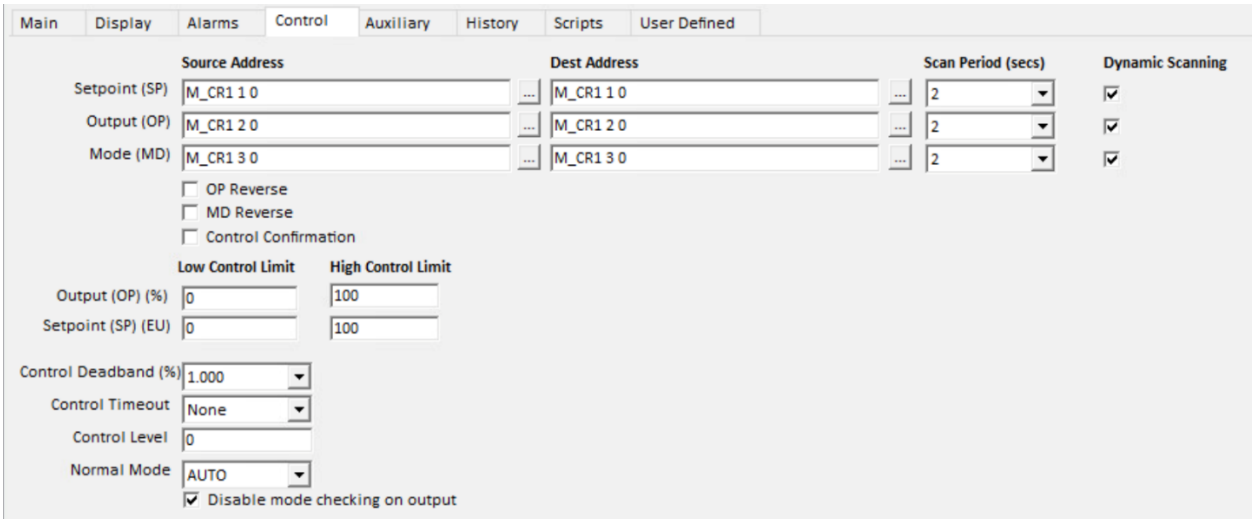


Figure 7 – Analog point configuration (Control tab)

On the History tab, I configured the parameters for long-term data storage. I enabled history collection not only for the Process Variable (PV) but also for the Setpoint (SP) and Output (OP). I also reviewed the Gate Point functionality, which allows the system to save database space by only gathering history when a specific "Gating Point" is in a defined state, rather than collecting data continuously.

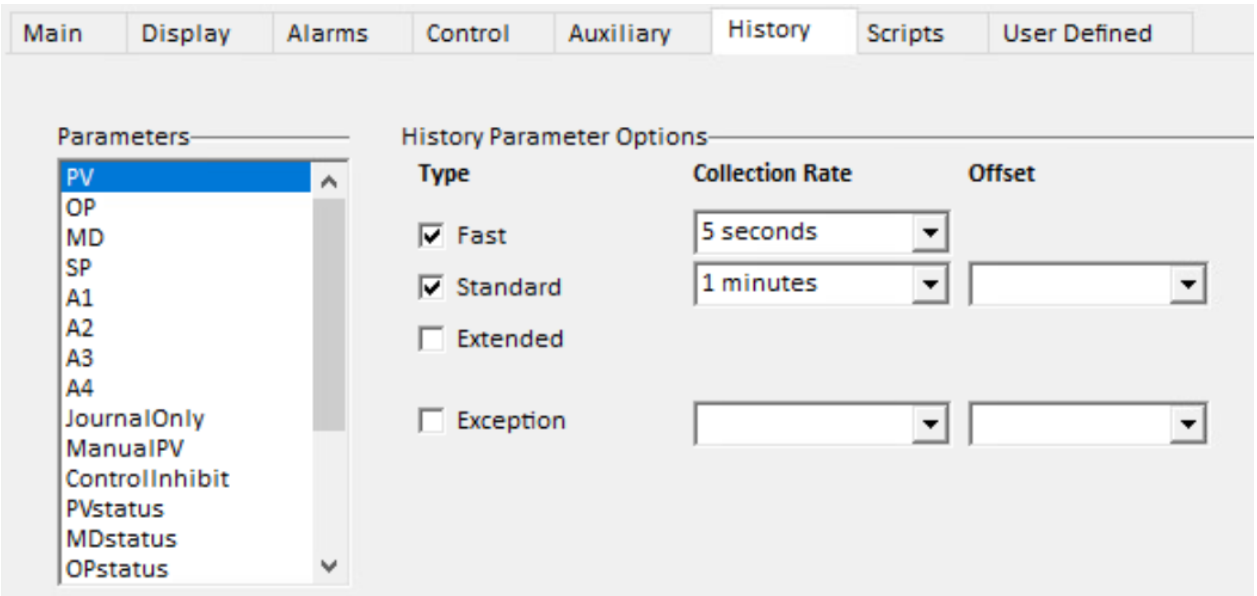


Figure 8 – Analog point configuration (History tab)

To add custom data storage to the server database, I navigated to the User Defined tab. I created a new parameter named USER_VAR and set its Link type to Variable. I defined the Value type as a 16-bit signed integer (INT2) with an initial value of 0. I specifically enabled the option to Update initial value from the server

during Upload to ensure any changes made during runtime are preserved during system synchronization.

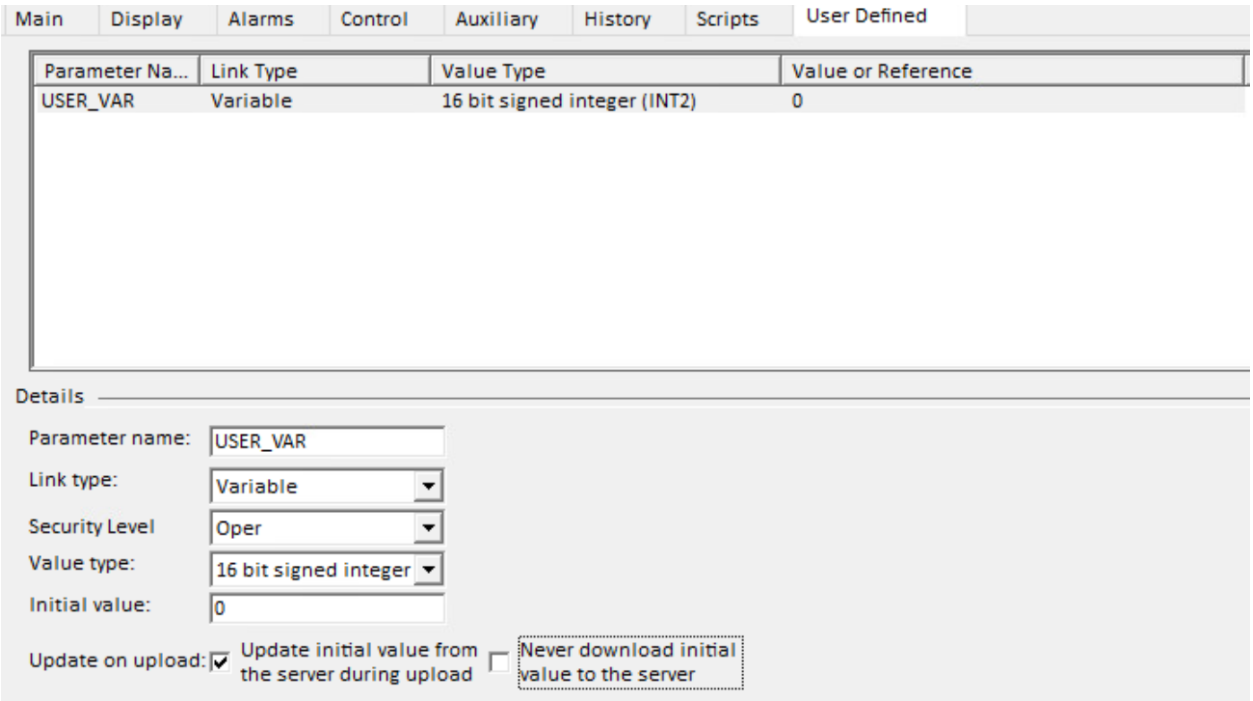


Figure 9 – Analog point configuration (User Defined tab)

After creating the user-defined variable, I returned to the History tab and located USER_VAR in the parameters list. I activated history collection for this specific parameter, ensuring that any changes to this custom integer value would be recorded alongside the standard analog data for future analysis.

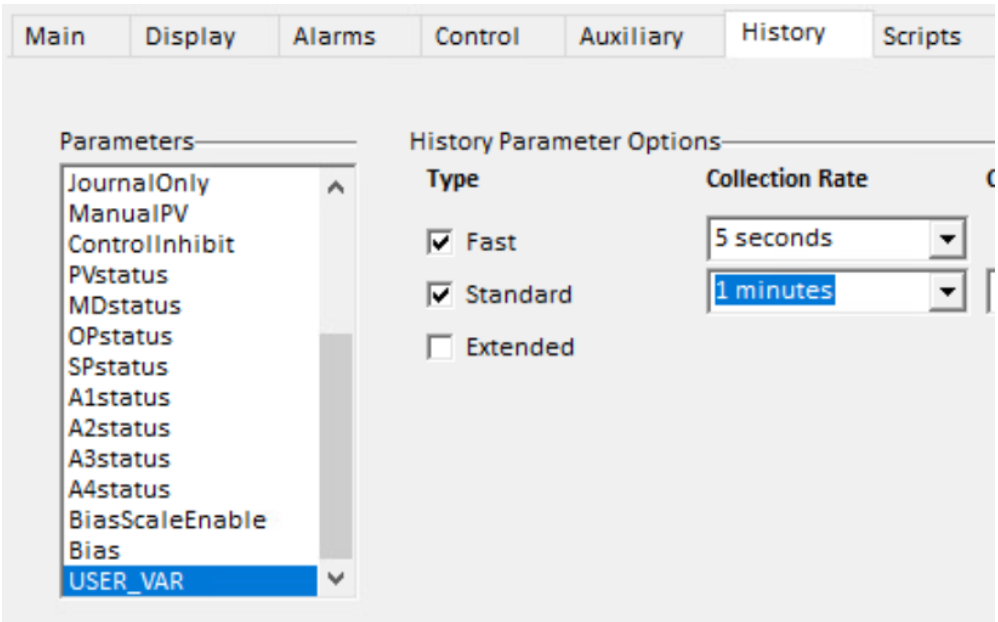


Figure 10 – Analog point configuration (History tab)

Once all tabs were fully configured, I clicked the Download icon to commit the changes to the Experion PKS server. In the download dialog, I confirmed that

Selected Items Only was highlighted and verified that the summary frame correctly identified 1 Analog point. I then executed the download and waited for the "OK" confirmation, signifying a successful database update.

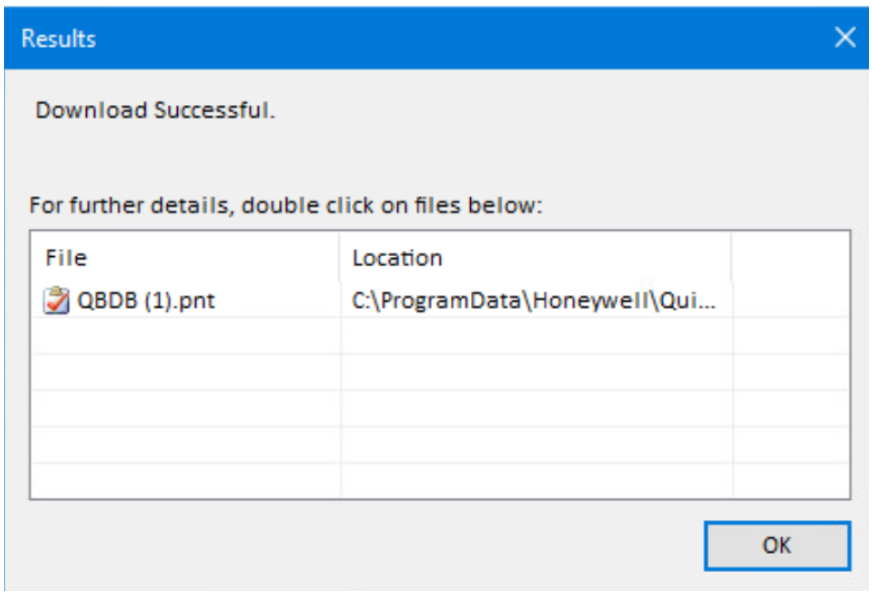


Figure 11 – After download results

Finally, I returned to the Main tab of the point details to verify the success of the operation. I checked the Last Modified and Last Downloaded fields to ensure they had been updated to the current date and time. This final check confirmed that the analog point was now active in the runtime environment and correctly communicating with the server.

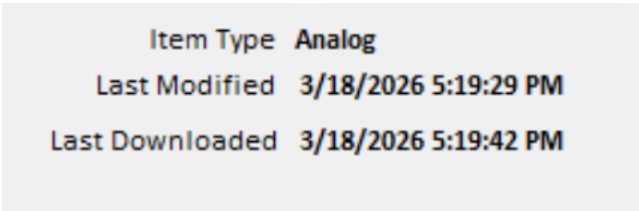


Figure 12 – Analog point status

Below is the current analog point, with alarm status: PV LoLow.

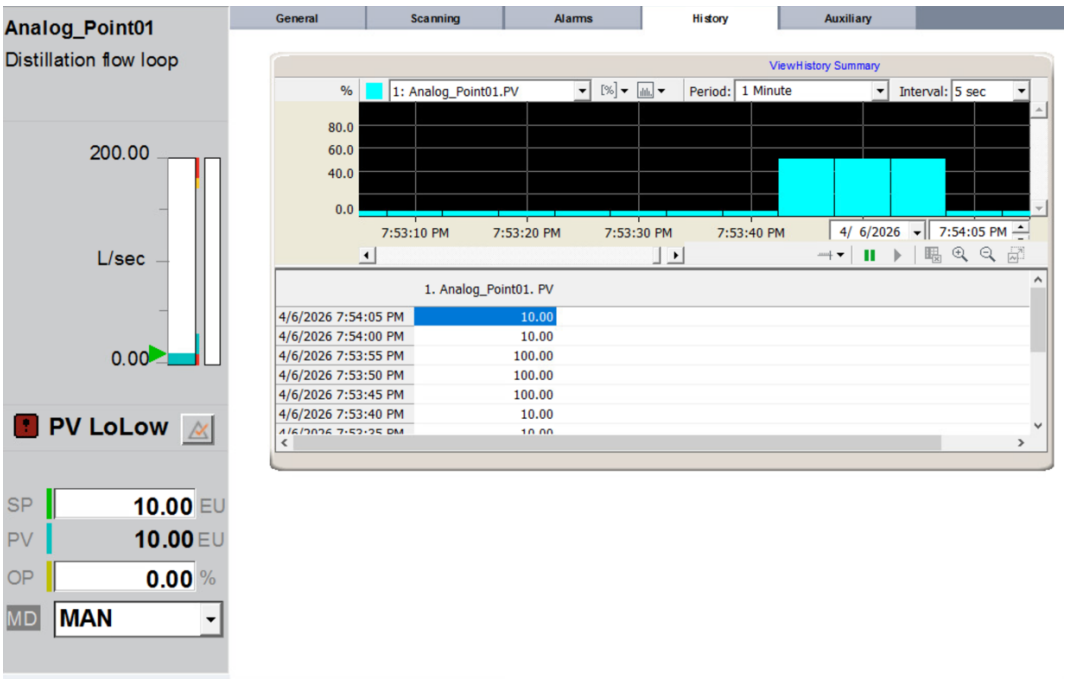


Figure 13 – Analog point in PV LoLow

Below is the active analog point, with alarm status: PV HiHigh.

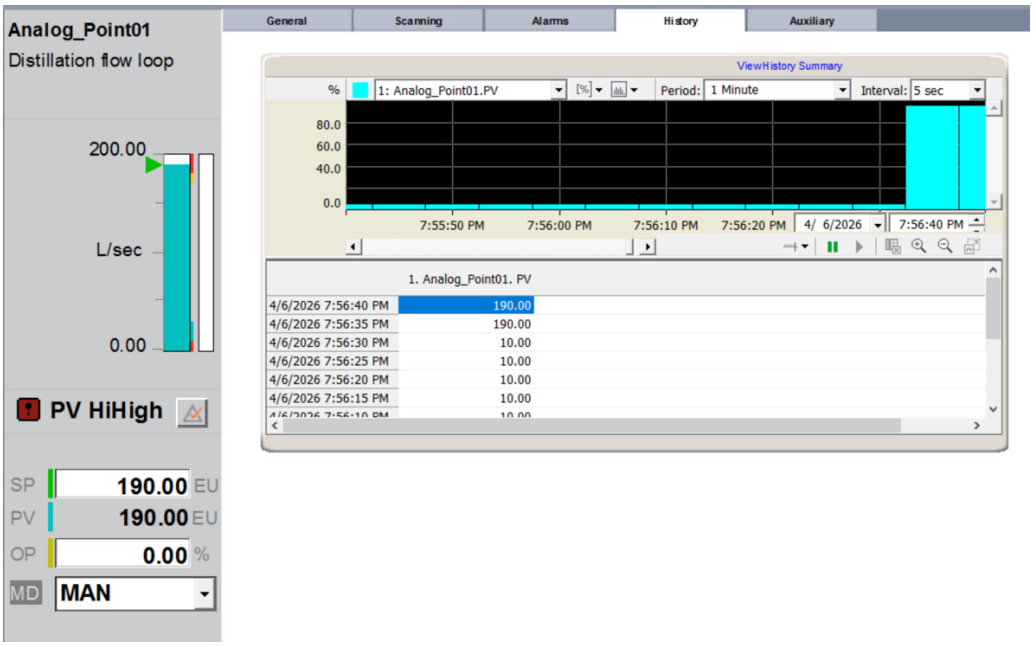


Figure 14 – Analog point in PV HiHigh

1.3 Task 3. Configure an accumulator point

To begin the final task of the laboratory, I initiated the creation of a new database point, selecting Accumulator Point from the type list. This point type is specifically designed to handle pulse-type data or totalizers from field devices.

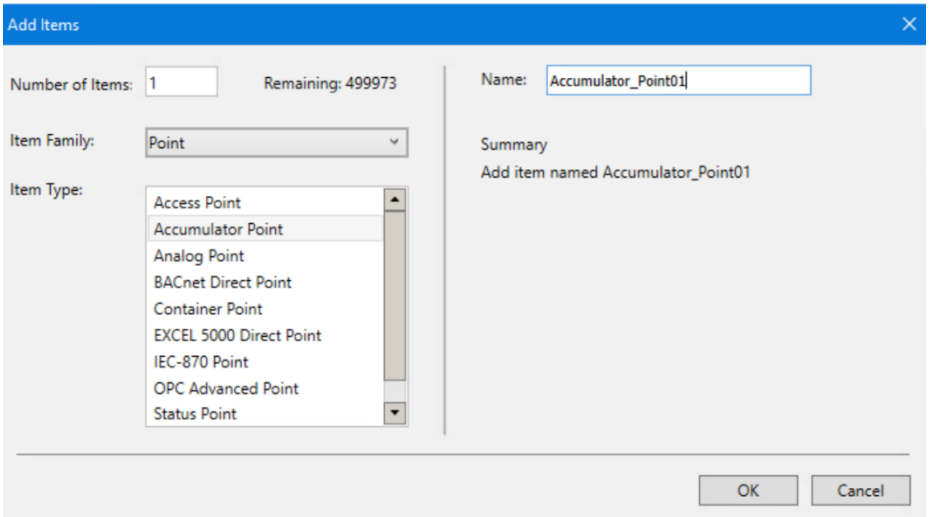


Figure 15 – Adding new accumulator point

On the Main tab, I entered "Distillation flow Loop" as the Description and linked the point to the Distillation_column asset. I configured the PV Source Address by selecting the M_CR1 controller and setting the location to 1 0. To ensure proper data processing, I set the Rollover Value to 4095, which defines the point at which the accumulator resets to zero.

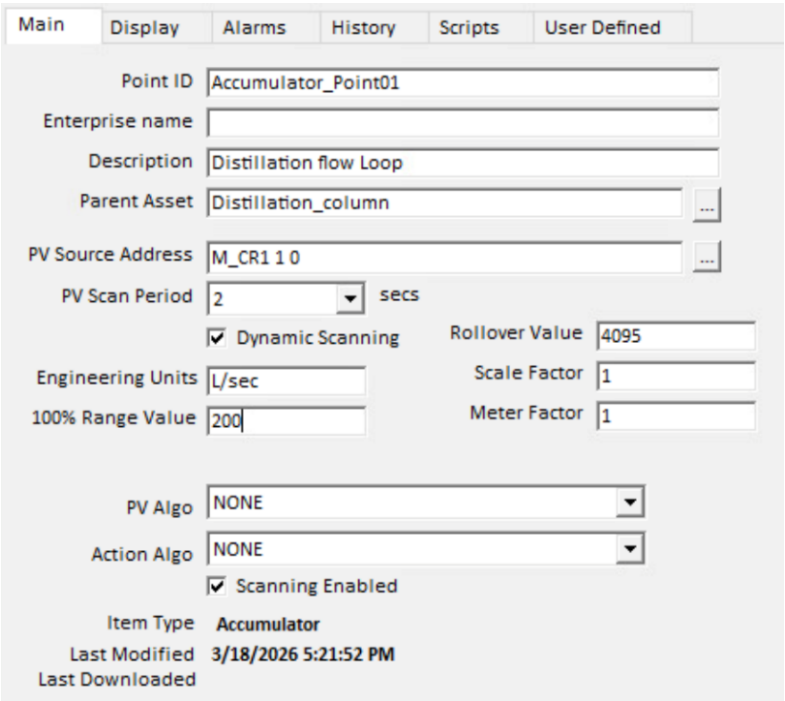


Figure 16 – Accumulator point configuration (Main tab)

I accessed the Display tab to integrate the accumulator into the monitoring environment, assigning it to a specific Group (1–16000) and Trend (1–3000) for visualization.

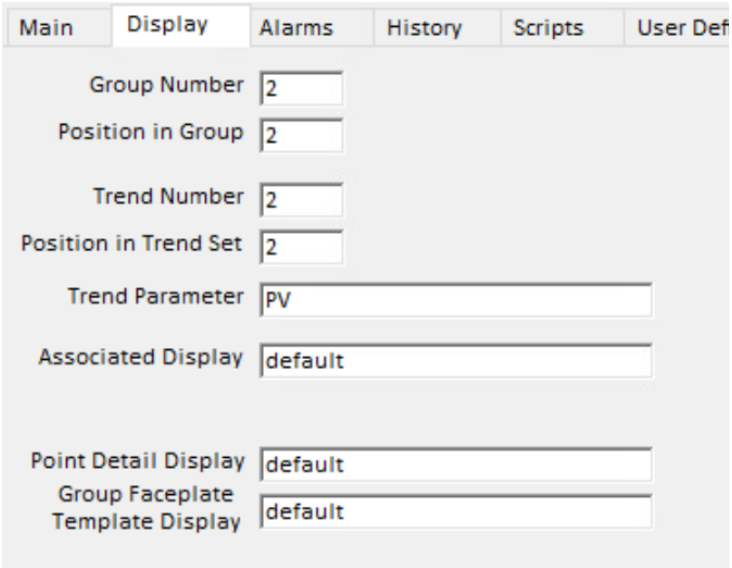


Figure 17 – Accumulator point configuration (Display tab)

Following this, I navigated to the Alarm tab. While accumulators often track totalized values, I configured the alarm settings to monitor the rate of accumulation or potential limit overflows, ensuring the operator receives timely notifications if the flow totals exceed expected operational parameters.

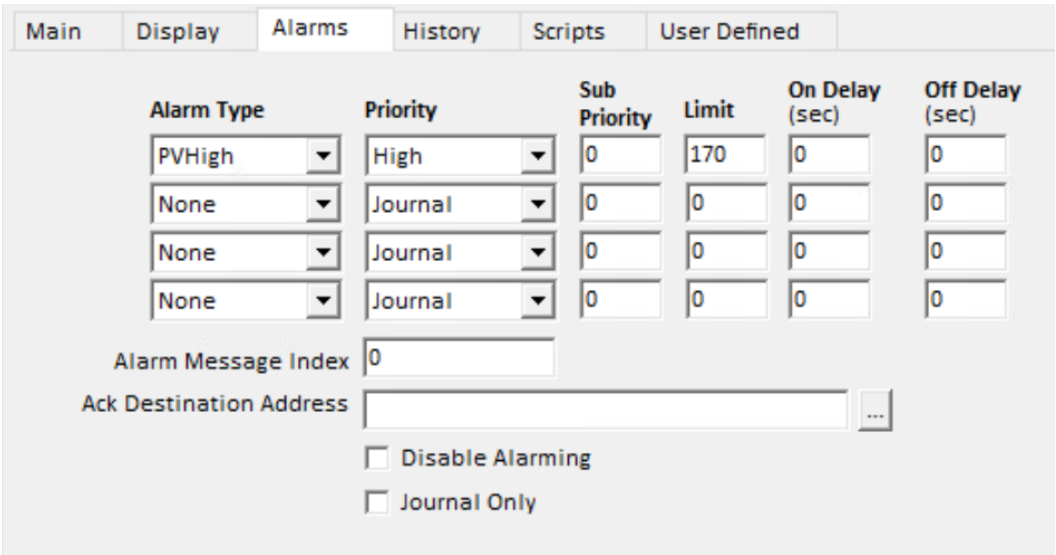


Figure 18 – Accumulator point configuration (Alarms tab)

On the History tab, I selected the PV (Process Variable) from the parameters list to enable continuous data collection. This is a critical step for an accumulator point, as it allows the system to store the increasing flow totals over time. By enabling history for the PV, I ensured that the server maintains a record of the

distillation flow, which can be used to generate consumption reports or volume-over-time trends.

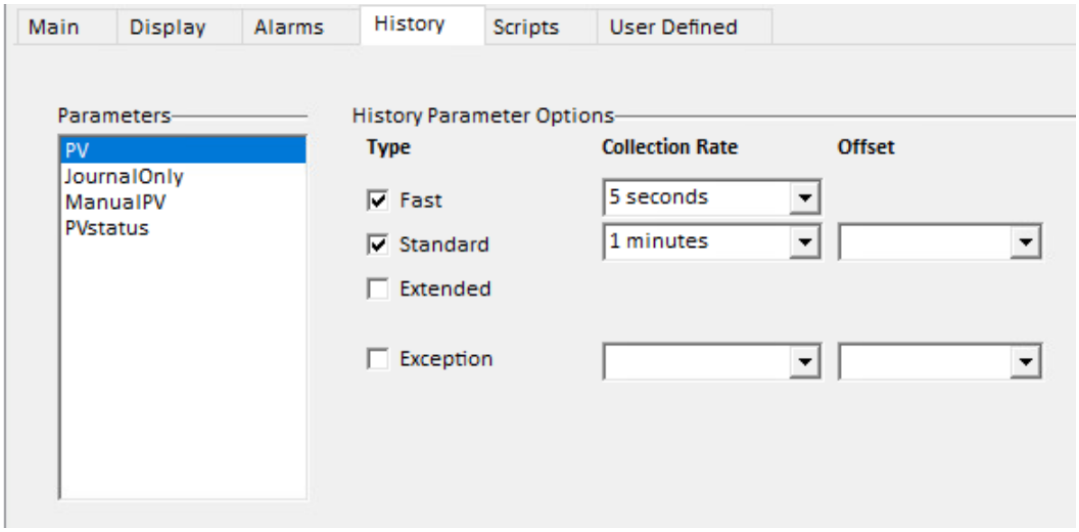


Figure 19 – Accumulator point configuration (History tab)

To enhance the point's functionality, I moved to the User Defined tab and created a new parameter named ACCUMULATOR_COUNT. I configured this as a Variable with a 16-bit signed integer (INT2) value type and an initial value of 0. I checked the Update initial value from the server during Upload box to ensure that if I ever need to upload the configuration back to Quick Builder, the most current count from the live server is retrieved and preserved.

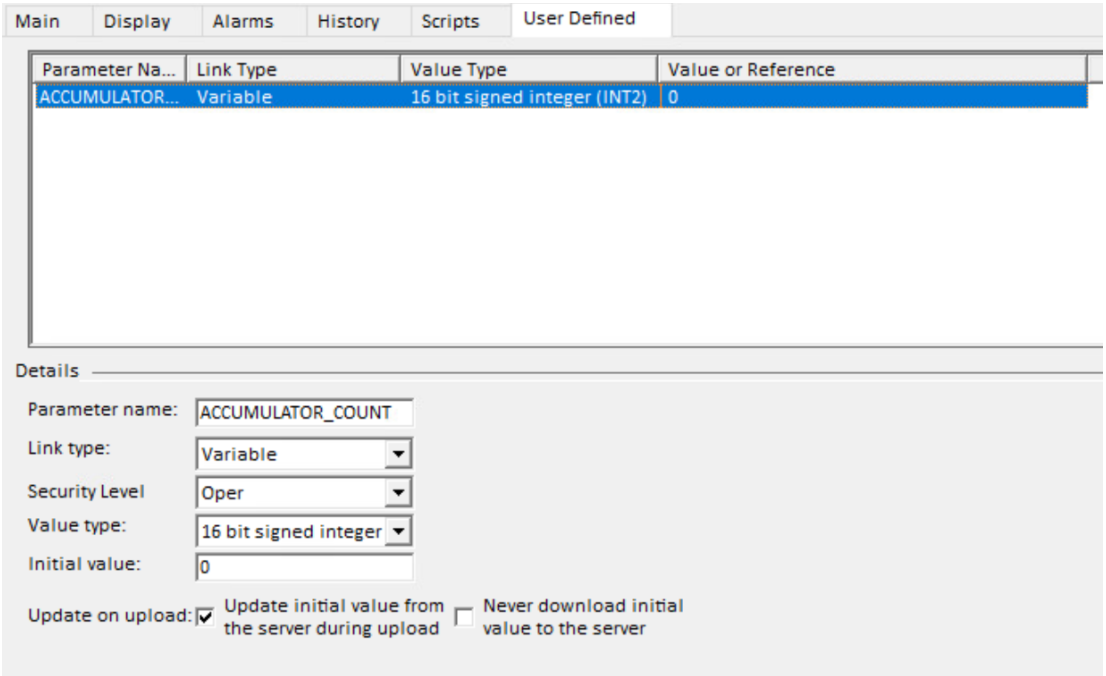


Figure 20 – Accumulator point configuration (User Defined tab)

Finally, I returned to the History tab once more to locate the newly created ACCUMULATOR_COUNT in the parameters list. I activated history collection for this specific user-defined variable.

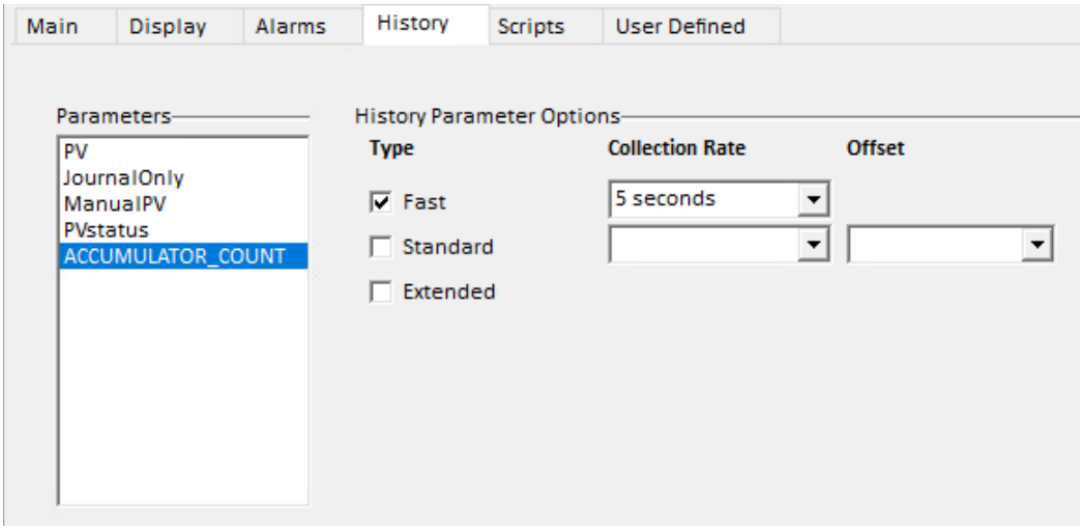


Figure 21 – Accumulator point configuration (History tab)

To commit the configuration of the Accumulator_Point01 to the system, I clicked the Download icon. After clicking Download, I received a confirmation message indicating a successful transfer of the point parameters to the Experion PKS server database.

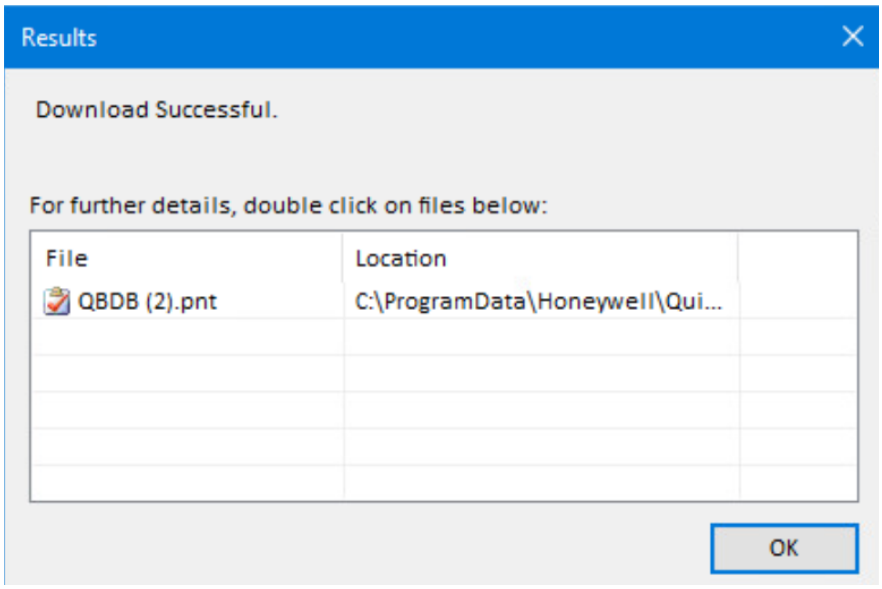


Figure 22 – After download results

As a final step, I returned to the Main tab of the accumulator point. I checked the Last Modified and Last Downloaded fields to confirm that the timestamps had been updated to the current date and time. This verification ensured that the server

was now actively tracking the distillation flow totals and that all communication parameters with the M_CR1 controller were properly synchronized and operational.

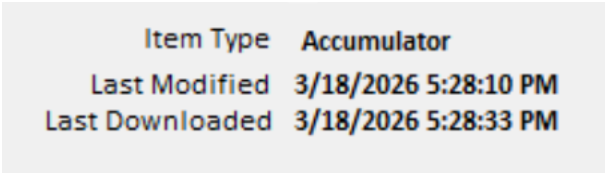


Figure 23 – Accumulator point status

Below is the current accumulator point, with alarm status: PV HiHigh.

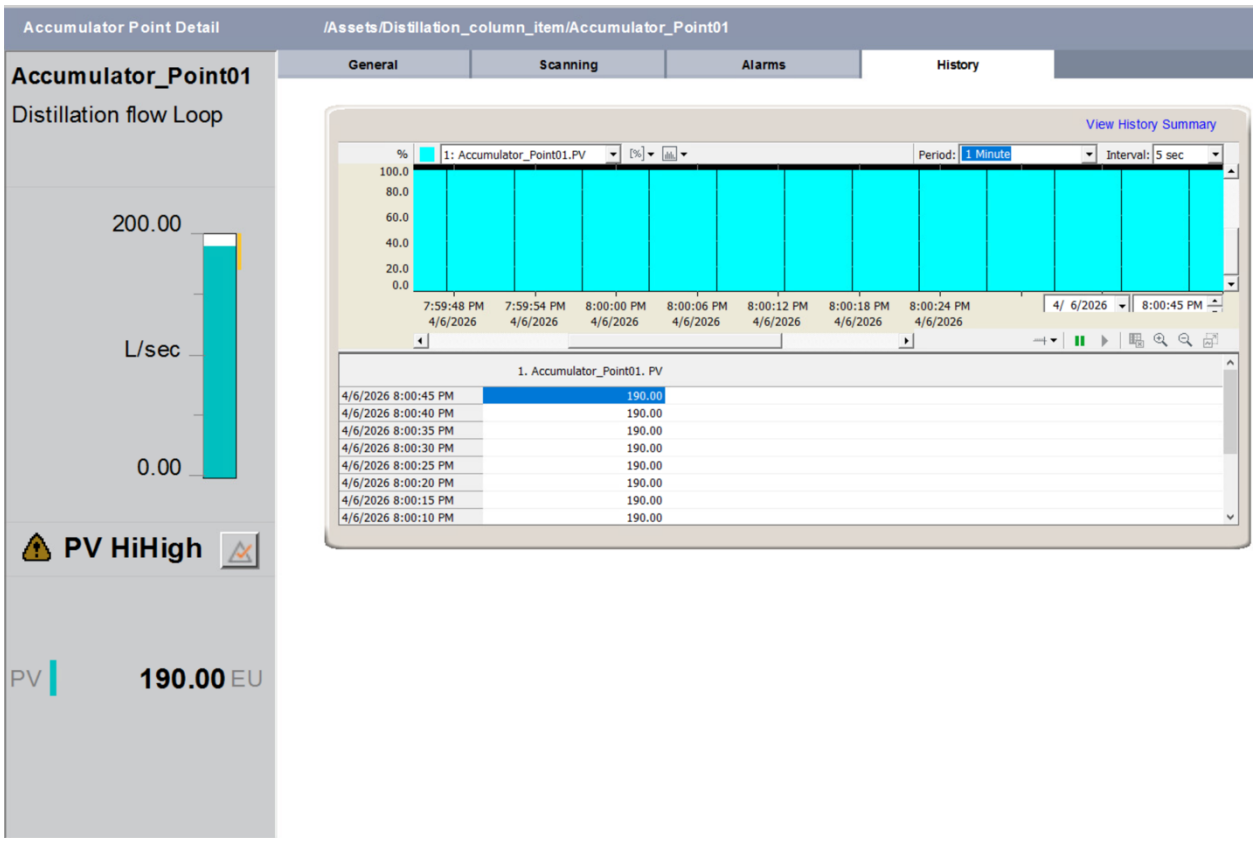


Figure 24 – Accumulator point in PV HiHigh

CONCLUSION

In conclusion, this laboratory work provided a comprehensive practical understanding of database engineering within the Honeywell Experion PKS environment. By utilizing the Quick Builder application, I successfully configured and deployed three distinct types of SCADA points—Analog, Status, and Accumulator—each serving a specific role in industrial process monitoring. This exercise demonstrated how raw data from M_CR1 and M_CR2 controllers is transformed into actionable information for operators, ensuring that continuous, discrete, and totalized variables are accurately represented in the system.

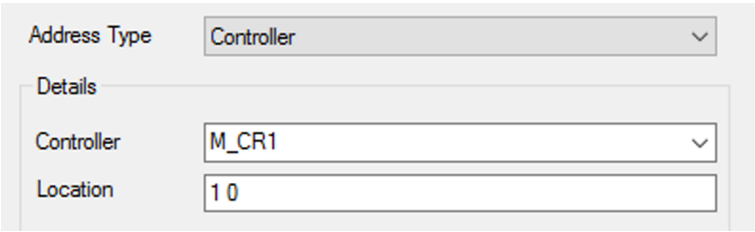
A critical takeaway from this work was the implementation of Alarm Management and History Collection. By setting hierarchical thresholds (such as PVHighHigh and PVLowLow) and defining priority levels, I learned how to ensure process safety through automated notifications. Furthermore, configuring history for both standard and User-Defined Parameters (USER_VAR, STATUS_COUNT) highlighted the importance of data archiving for long-term trend analysis and diagnostic troubleshooting, which are essential for maintaining operational efficiency in a real-world plant.

Finally, the laboratory emphasized the necessity of Database Synchronization through the Download process. Verifying the Last Modified and Last Downloaded timestamps confirmed that the logical configurations were correctly active in the server's runtime environment. Overall, this task solidified the technical skills required to build a robust SCADA database, bridging the gap between field-level hardware and high-level supervisory control, which is a fundamental requirement for any automation engineer.

WORKSHEET

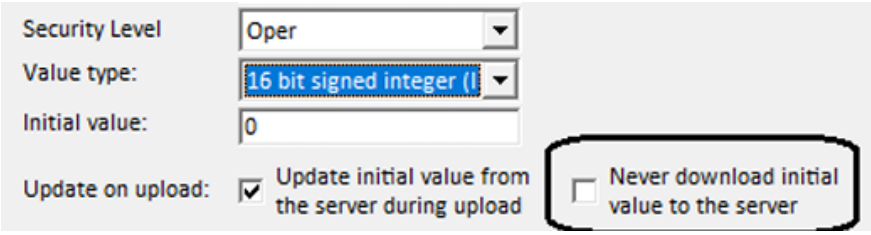
Student name: Fazkullin Farid, Zhanshin Temirlan

1. **(25 points)** What does the numbers of Location represent?



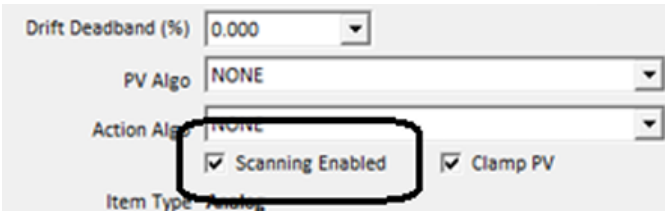
The numbers in the Location field define the specific memory address within the controller where the point's data is stored. The first integer (e.g., "1") represents the address of the Holding Register, which is the primary data storage unit in the controller's memory. The second integer (e.g., "0") specifies the bit number within that register.

2. **(25 points)** What is the purpose of this checkbox?



The "Never download initial value to the server" checkbox is a safety feature used to protect current runtime data. When this box is selected, the initial value defined in the Quick Builder configuration will not be sent to the server database during the download process.

3. **(25 points)** What is the purpose of this checkbox?



The "Scanning Enabled" checkbox serves as the activation switch for real-time data communication between the server and the field controller. When this checkbox is selected, it instructs the Experion PKS server to begin refreshing the point's database by continuously reading data from the assigned controller registers.

4. **(25 points)** What does mean an Analog point?

Analog Point is a fundamental database entity in Experion PKS used to represent and process continuous variables. Unlike status points, which represent discrete "on/off" states, an analog point handles data that varies over a numerical range, such as temperature, pressure, or flow rate (e.g., 0 to 200 L/sec).